Sharzil Hasnine

+61 451 645 516 | sharzilhasnine975@gmail.com | linkedin.com/in/sharzilhasnine | github.com/Shas-9

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Hiring Manager
ProFound – People in Technology
Adelaide, SA

Re: Software Engineer (Rust, C++)

Dear Hiring Manager,

I am writing to apply for the Software Engineer role with the company you're recruiting for. As a Computer Science (Advanced) student at Adelaide University with strong, hands-on C++ experience and real exposure to robotics and hardware-adjacent systems, I am excited by the chance to work close to the physical world rather than on another SaaS dashboard.

My C++ experience runs deep. In an industry project with Maptek, I architected a stream-oriented lossless compression system for 3D data entirely in C++, achieving a 98.8% reduction in data size and a 2.6x processing-speed improvement, backed by a purpose-built testing framework and a cross-platform CI pipeline with reproducible Linux and Windows builds. I also built the Paxos Consensus Algorithm from scratch in Java with a TCP socket-based messaging protocol, and a 2D game in C++ and SFML with a modular, object-oriented architecture separating game state, entities, and UI – so I am comfortable designing and reasoning about systems software, not just scripting on top of them. Alongside this, I have direct experience working close to hardware: as a Technical Volunteer with the Adelaide Rover Team, I designed a behavior tree for a mobile rover's autonomous navigation pipeline in C++ and Python, and validated it through simulation ahead of the Australian Rover Challenge, working directly alongside robotics and mechatronics-focused teammates.

I have not yet worked in Rust, and I want to be upfront about that rather than overstate it. What I can point to instead is a track record of picking up new languages and tools quickly and applying them properly under real constraints – ROS2, Nav2, and Gazebo for the rover project, and distributed-systems fundamentals (Lamport clocks, consensus protocols) implemented from first principles without external libraries. Given my depth in C++ and comfort with low-level, correctness-focused engineering, I am confident I would ramp up on Rust quickly and productively.

I use Git as a matter of course across every project, value testing and CI/CD as core engineering practice rather than an afterthought, and would welcome the ownership and high-trust environment this role offers. I'd be glad to discuss my background further.

Thank you for your time and consideration.

Sincerely,
Sharzil Hasnine